DVA CAPSTONE 2

WHY STARTUPS FAIL

VC Investment Pattern Analysis

Crunchbase Dataset · 49,437 Startups · 1990–2014

Sector: Finance / Venture Capital / Startup Ecosystem

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1. Executive Summary

Written for: VC fund partners and investment committee members. No technical background required.

Problem. Venture capital firms deploy hundreds of billions annually into early-stage companies, yet a significant share is permanently lost when startups fail. Fund managers rely on intuition and pattern-matching with no systematic, data-backed framework to flag high-risk candidates before capital is deployed.

Approach. Raw Crunchbase investment data (54,294 rows, 39 columns) was processed through a 7-step Python ETL pipeline, yielding 49,437 cleaned startup records. Eight EDA visualisations and five statistical tests — Mann-Whitney U, Chi-square, Kruskal-Wallis, and logistic regression — produced the findings below, published in an interactive Tableau dashboard with executive and operational views.

Key Insights.

- Closed startups raised a median \$1.0M vs \$1.76M for operating firms — underfunding is the strongest measurable predictor of failure ($p < 0.001$).
- Single-round startups fail at the highest rate of any cohort — no follow-on investor is a critical warning signal.
- Sector failure rates are statistically uneven (Kruskal-Wallis, $p < 0.001$); consumer-facing internet sectors fail at 2× the rate of enterprise software.
- The 2006–2008 founding cohort produced nearly double the average failure rate, confirming vintage-year macro conditions amplify structural risk.

Key Recommendations.

1. Flag all single-round portfolio companies for a 90-day review before deploying follow-on capital.
2. Require co-investor commitment for any seed deal below \$500K; raise to \$750K for high-risk sectors.
3. Apply sector-adjusted return hurdles: ≥5× for consumer-app/curated-web categories; ≥3× for enterprise and infrastructure.

2. Sector & Business Context

Sector overview. The VC and startup ecosystem is one of the most information-rich yet analytically underserved segments of global finance. Despite decades of outcome data, most allocation decisions are still made on qualitative judgment and surface-level pattern-matching. The dataset spans 1990–2014, covering three major economic cycles: the dot-com expansion and crash (1995–2002), the Web 2.0 recovery (2003–2007), and the global financial crisis and aftermath (2008–2012) — providing a natural test of whether failure signals hold across radically different macro environments.

Decision-maker. This analysis is designed for **early-stage VC fund managers, portfolio analysts, and investment committee members** at firms deploying \$20M–\$500M per fund. These practitioners screen 200–500 companies per year and make 10–30 investment decisions per cycle. One missed failure signal can represent \$250K–\$1M+ in unrecoverable capital.

Business value. A fund deploying \$50M across 100 companies at \$500K each expects ≈\$2.5M in lost capital at the 5.27% observed failure rate. Reducing that rate by even 1 percentage point saves \$500K per cycle. Existing research focuses on academic survival analysis; this project translates findings into an investor-facing, actionable screening framework.

3. Problem Statement & Objectives

Core Business Question: Which funding, sector, geographic, and timing factors are statistically significant predictors of startup failure — and what actionable thresholds can investors use to improve capital allocation decisions?

Project Scope.

In Scope	Out of Scope
Historical analysis of 49,437 startup records	Real-time scoring or model deployment
EDA across funding, sector, geography, time	Post-2014 data; qualitative factors
Statistical hypothesis testing ($\alpha = 0.05$)	Individual company-level recommendations
Logistic regression + Tableau dashboard	

Success Criteria.

- $\geq$5 statistically validated findings ($p < 0.05$) linking measurable attributes to closure.
- A logistic regression model with significant coefficients for key predictors.
- An interactive Tableau dashboard with executive and operational views.
- 3–5 actionable, quantified business recommendations.

4. Data Description

Table 1: Dataset Overview

Attribute	Details
Source	Crunchbase Startup Investments (Kaggle) — dataset link
Raw / Cleaned	54,294 rows $\rightarrow$ 49,437 rows (4,857 duplicates removed)
Columns	39 raw $\rightarrow$ 42 post-ETL (including 6 engineered fields)
Time Period	1990–2014
Unit	One row = one startup entity

Table 2: Key Column Definitions

Column	Type	Description	Role in Analysis
status	String	operating/closed/acquired/ipo	Source of target variable
funding_total_usd	Float	Total USD funding received	Primary continuous predictor
funding_rounds	Integer	Number of distinct rounds	Count predictor
market	String	Primary sector/industry label	Categorical segmentation
country_code	String	ISO 3-letter country code	Geographic filter
is_closed*	Binary	1 = closed, 0 = other	Target variable
funding_tier*	String	Micro/Seed/Growth/Late Stage	Segmentation variable
reached_series_a*	Binary	1 if round_A > 0	Milestone indicator
log_funding*	Float	log(1 + funding_usd)	Normalized regression input

*Engineered during ETL pipeline.

Known Data Limitations:

- **Survivorship bias:** Startups without a digital footprint may be absent from Crunchbase, understating true ecosystem failure rates.
- **Status lag:** Cohorts founded 2010–2014 listed as "operating" may have since closed — outcomes require time to materialise.
- **Geographic skew:** USA accounts for ≈60% of records; non-US conclusions carry wider uncertainty.
- **Funding gaps:** 17.26% of `funding_total_usd` is missing — zero values treated as undisclosed, not literal zeros.
- **Self-reported sectors:** Market labels may be inconsistent across similar companies.

5. Data Cleaning & ETL Pipeline

All cleaning was performed in Python (`pandas`, `numpy`), fully reproducible via `notebooks/02_cleaning.ipynb` and `scripts/etl_pipeline.py` on GitHub.

Table 3: ETL Pipeline — Step-by-Step Log

#	Description	Before	After	Dropped
1	Load raw data, standardize column names	54,294	54,294	0
2	Clean text noise, normalize missing tokens	54,294	54,294	0
3	Remove exact and permalink duplicates	54,294	49,437	**4,857**
4	Parse and validate all date fields	49,437	49,437	0
5	Standardize numeric funding fields	49,437	49,437	0
6	Derive engineered analysis features	49,437	49,437	0
7	Export cleaned analytical dataset	49,437	49,437	0
Total reduction		54,294	**49,437**	**4,857 (8.95%)**

Key assumptions and design decisions:

- Acquired and IPO outcomes treated as positive exits (`is_closed = 0`).
- No aggressive outlier deletion — VC funding distributions are inherently heavy-tailed.
- Duplicates resolved by retaining the first record per `permalink`.
- Missing subsets excluded only from their respective statistical tests.

6. KPI & Metric Framework

All KPIs computed in `notebooks/04_statistical_analysis.ipynb` and exported to `data/processed/kpi_summary.csv`

Table 4: KPI Framework — Values and Business Mapping

KPI	Value	Business Link
Overall Failure Rate	**5.27%**	Baseline risk benchmark for portfolio planning
Funding Gap (op. vs closed)	**$757,977.5**	Minimum capital adequacy gap to track
Median Funding — Closed	**$1,000,000**	Underfunding danger threshold
Median Funding — Operating	**$1,757,977.5**	Survival capital benchmark
Avg Rounds — Closed	**1.43**	Round-count danger threshold
Avg Rounds — Operating	**1.69**	Follow-on investor signal
Avg Rounds — Acquired	**2.01**	Acquisition capital profile
Operating Share	**84.61%**	Portfolio-state context
Acquisition Share	**7.47%**	Positive-exit benchmark (separate from failure)
Series A Reach Rate	**18.21%**	Milestone progression signal

7. Exploratory Data Analysis

All EDA performed in `notebooks/03_eda.ipynb`; charts saved to `tableau/screenshots/`.

7.1 Status Distribution

84.61% of startups are operating, 5.27% permanently closed, and 7.47% acquired. The small failure class means any predictive model must use precision/recall metrics — not accuracy — to avoid the majority-class trap. *(eda_01)*

7.2 Failure by Funding Rounds — Single Round Is the Danger Zone

Failure is highest for single-round startups and declines monotonically as rounds increase. Repeat investor participation signals company health beyond capital volume alone — a startup that has raised two rounds has been market-vetted twice. *(eda_02)*

7.3 Failure by Market Sector

Table 5: Top Sectors by Failure Rate (min. 50 startups)

Sector	Failure Rate	Risk Level
Public Relations	20.47%	**Very High**
Facebook Applications	16.90%	**High**
Location Based Services	16.67%	**High**
iPhone	15.62%	**High**
Curated Web	15.52%	**High**
Social Network Media	14.04%	**Moderate**
Shopping / Music	12.50%	**Moderate**
Enterprise Software	≈7%	**Lower**
Biotechnology	≈8%	**Lower**

7.4 Additional EDA Findings

- **Funding distribution** *(eda_04):* Closed-startup funding curve is visibly left-shifted. Median gap $757,977.5 confirmed by Mann-Whitney U ($p < 0.001$).
- **Failure trend** *(eda_05):* 2007 is the peak failure cohort, aligning with pre-crisis credit expansion followed by sudden capital contraction.

- **Country risk** *(eda_06):* Malaysia (12.50%), Estonia (11.63%), New Zealand (10.17%) show elevated rates. USA closure: 5.18% vs 4.94% non-USA.
- **Correlation heatmap** *(eda_07):* No single variable exceeds $|r| = 0.06$ with closure — confirming multi-causal failure structure.
- **Series A survival** *(eda_08):* Chi-square ($\chi^2 = 576.41$, $p < 0.001$, Cramér's $V = 0.108$) — significant but moderate effect; one signal among several.

8. Statistical Analysis

All tests run in `notebooks/04_statistical_analysis.ipynb` at $\alpha = 0.05$.

Test 1 — Mann-Whitney U: Funding vs. Closure Status

H_0: Funding distributions for closed and operating startups are equal.
$U = 32{,}474{,}908, \quad p < 0.001 \quad \Rightarrow$ **Reject** H_0

Closed startups raise significantly less capital. Underfunding is a structural, repeatable predictor — not noise.

Test 2 — Chi-Square: Series A Achievement vs. Outcome

H_0: Series A achievement is independent of startup outcome.
$\chi^2 = 576.41, \; df = 3, \; p < 0.001, \;$ Cramér's $V = 0.108 \quad \Rightarrow$ **Reject** H_0

Significant but moderate effect — Series A is one important signal, not a standalone survival guarantee.

Test 3 — Kruskal-Wallis: Funding Rounds by Outcome

H_0: Round distributions are equal across all status groups.
$H = 519.52, \quad p < 0.001 \quad \Rightarrow$ **Reject** H_0
Acquired: 2.01 | Operating: 1.69 | Closed: 1.43

Sustained follow-on investor participation is statistically linked to better outcomes.

Test 4 — Kruskal-Wallis: Sector-Level Failure Differences

H_0: Failure rates are equal across major sectors. $\quad p < 0.001 \quad \Rightarrow$ **Reject** H_0

Public Relations (20.47%) vs Enterprise Software (≈7%). Sector selection materially changes the risk a fund is carrying — return hurdles cannot be uniform.

Test 5 — Logistic Regression: Independent Predictors of Closure. Model fitted on 31,437 complete-case startups. Target: `is_closed`.

Table 6: Logistic Regression Results

Variable	Odds Ratio	P-value	Interpretation
funding_rounds	0.764	< 0.001	Each extra round reduces closure odds ≈24%
log_funding ($\log_{10}$)	0.912	< 0.001	Higher funding reduces odds after controls
days_to_first_funding	0.9998	< 0.001	Earlier first funding is directionally protective
reached_series_a	1.207	0.0096	Positive after controls — likely selection effects
is_usa	1.093	0.0915	*Not significant at 5%*

Both funding path (rounds) and funding volume independently reduce closure odds. A startup raising across multiple rounds is measurably safer than one single raise of equal total.

Point-biserial correlations confirm no individual predictor exceeds $r = 0.06$ with `is_closed` — failure is multi-causal and requires a composite screening framework.

9. Tableau Dashboard Design

Objective. Enable VC partners to screen investment candidates against historical failure signals and monitor existing portfolio holdings against data-backed risk benchmarks — in a single, interactive view.

Table 7: Dashboard View Structure

View	Components	Audience
Executive	Failure rate KPI tile; funding gap card ($1.0M vs $1.76M); status distribution donut; year-over-year failure rate trend	Fund partners, investment committee
Operational	Sector failure rate bar chart (ranked, filterable by decade); country heatmap; funding scatter plot; Series A survival comparison	Portfolio analysts, deal teams

Interactive Filters: Country (multi-select) · Market/Sector (multi-select) · Founding Decade (dropdown) · Status (multi-select) · Funding Rounds (range slider).

Live Dashboard URL:

https://public.tableau.com/views/tableau_17774462589810/Dashboard1

Published on Tableau Public — accessible without login. Use filters on the right panel to explore sector, country, and funding-round risk dimensions.

10. Insights Summary

1 — Underfunding Is the Strongest Measurable Predictor of Failure

Closed startups: median $1.0M vs $1.76M for operating firms — a $757,977.5 gap confirmed by Mann-Whitney U ($p < 0.001$). Any portfolio company below this threshold warrants immediate capital review.

2 — Single-Round Startups Are the Highest-Risk Cohort

Receiving only one funding round with no follow-on is the clearest high-risk signal available at the portfolio monitoring stage. Every single-round company warrants proactive outreach.

3 — Series A Is Significant, But Not a Survival Guarantee

Cramér's $V = 0.108$ confirms the association is real but moderate. Series A should be tracked as one flag within a multi-signal composite, not used as a binary pass/fail gate.

4 — Consumer Internet Sectors Carry 2× the Baseline Failure Rate

Public Relations (20.47%), Facebook Applications (16.90%), and Curated Web (15.52%) require materially higher expected return multiples to compensate for elevated closure risk.

5 — The 2006–2008 Cohort Produced Nearly Double the Average Failure Rate

Pre-crisis credit expansion followed by sudden capital contraction created a peak failure vintage. Macro conditions amplify structural funding weaknesses — vintage-year risk is real and measurable.

6 — The Funding Path Matters as Much as the Amount

Logistic regression: funding_rounds ($OR = 0.764$, $p < 0.001$) and log_funding ($OR = 0.912$, $p < 0.001$) both *independently* reduce closure odds. Multiple rounds > one large single raise of equal size.

7 — Geographic Risk Is Real but Context-Dependent

Smaller ecosystems (Malaysia, Estonia, New Zealand) show elevated failure rates. Geography should be used as a composite modifier, not a standalone rejection signal.

8 — Failure Is Concentrated and Predictable, Not Random

Five statistical tests jointly confirm failure clusters around measurable, observable attributes available at the time of investment. A composite screening framework is feasible at zero additional data-collection cost.

9 — Acquisition Is Not Failure — Separate KPIs Are Required

7.47% of the dataset was acquired — a positive exit. Track the closed rate (5.27%) and acquired rate separately; conflating them understates the fund's true positive-exit performance.

10 — No Single Variable Predicts Failure — A Composite Framework Is Essential

No individual predictor exceeds $|r| = 0.06$ with closure. Applying any single threshold will systematically miss companies that are well-funded but single-round and operating in a high-failure sector.

11. Recommendations

Rec 1 — Round-Count Screening Priority: **High** | Insight 2

Flag all single-round portfolio companies for a mandatory 90-day intervention assessment before any follow-on capital is deployed.
Expected impact: 10–15% reduction in portfolio failure rate → \$500K–\$750K saved per \$50M fund cycle.

Rec 2 — Capital Adequacy Floor Priority: **High** | Insight 1

Require co-investor commitment for any seed deal below \$500K; raise floor to \$750K for consumer-app and curated-web sectors.
Expected impact: Shifts portfolio median funding above the \$1.0M danger threshold.

Rec 3 — 18-Month Series A Milestone Clock Priority: **Medium** | Insight 3

Assign analyst support, warm introductions, and bridge capital options for companies approaching 18 months without a Series A term sheet.
Expected impact: Converting one additional company per cycle reduces expected failures by ≈0.5 pp.

Rec 4 — Sector-Adjusted Return Hurdles Priority: **Medium** | Insight 4

Require ≥5× target multiple for Public Relations, Facebook Applications, Curated Web; ≥3× for enterprise and infrastructure.
Expected impact: Prevents systematic underpricing of risk at investment committee approval stage.

Rec 5 — Macro-Sensitivity Monitoring Priority: **Low** | Insight 5

Track cohort failure rates against credit spreads and VC fundraising volumes; reduce deal velocity 20–30% when two+ macro indicators signal contraction.
Expected impact: 15–25% reduction in crisis-cohort failure exposure.

12. Impact Estimation

Table 8: Fund-Level Capital Savings Across Scenarios (Conservative 1 pp Failure Rate Reduction)

Fund	Size	Portfolio	Current Failures	Post-Framework	Saved
Small	\$20M	40 @ \$500K	≈2 = \$1.0M	≈1.7	\$150K–\$300K
Mid	\$50M	100 @ \$500K	≈5 = \$2.5M	≈4.3	\$350K–\$750K
Large	\$200M	200 @ \$1M	≈10 = \$10.0M	≈8.6	\$1.4M–\$2.1M

Cost of inaction. Without a data-backed screening layer, fund managers will continue to over-invest in single-round, undercapitalised companies that 25 years of historical data consistently show close at the highest rates. Every signal identified requires **no new data collection and no technology investment** — the information is already available in existing Crunchbase and pitch deck records.

13. Limitations

- **Causation cannot be inferred.** Underfunding may be a symptom of poor fundamentals, not a direct cause — association is identified, not mechanism.
- **Failure rate is likely understated.** Survivorship bias and status lag mean 5.27% is a floor estimate.
- **Individual company predictions are not possible.** This framework provides population-level risk signals, not binary go/no-go for specific companies.
- **Acqui-hire vs strategic acquisition** are not distinguished — both treated as positive exits.
- **Missing funding values** (17.26%) introduce selection bias toward startups with disclosed funding.
- **Self-reported sector labels** may be inconsistent, particularly at sector intersections.

14. Future Scope

Additional analysis.

- **Survival analysis** (Kaplan-Meier, Cox model): model time-to-failure rather than binary outcome.
- **ML classifier** (Random Forest, XGBoost): per-startup failure probability scores for deal screening.
- **Segmented regression:** separate logistic models per sector and decade to test coefficient stability.

Additional data sources.

- **Post-2014 PitchBook data:** validates whether signals hold in the AI/SaaS era.
- **Founder background data** (prior exits, repeat-founder status): human capital predictors alongside financial ones.
- **VC firm-level data** (fund size, co-investor network quality): tests whether investor identity predicts outcomes.

What can be built next.

- **Predictive scoring API:** REST endpoint returning a failure probability score integrable into deal management software.
- **Live Tableau pipeline:** connect the dashboard to a Crunchbase API feed for continuous portfolio monitoring.

- **Sector risk composite index:** weighted risk score combining failure rate, funding gap, and avg round count, updated quarterly.

15. Conclusion

Can startup failure be predicted from observable signals available at the time of investment? Based on 49,437 cleaned Crunchbase records and five rigorous statistical tests — **yes, meaningfully**.

The analysis identified a consistent, statistically validated failure profile: closed startups are characterised by lower total funding (median $1.0M vs $1.76M for operating firms), fewer funding rounds (1.43 vs 1.69), weaker milestone progression, and concentration in consumer-facing sectors during adverse macro conditions. None of these signals are hidden — they exist in current portfolio data at the time of every investment decision.

The recommended action is immediate and low-cost: implement a composite screening framework that flags single-round, underfunded candidates before capital is deployed and monitors pre-Series A companies against an 18-month milestone clock. Conservative estimates place savings at **$350K–$750K per $50M fund cycle**. No new data. No new technology. The signals are already there.

A. Data Dictionary — Full Column Definitions

Table 9: Key Columns — Raw and Engineered

Column	Type	Formula / Source	Missing %	Notes
status	String	Raw Crunchbase	2.66%	operating/closed/acquired/ipo
funding_total_usd	Float	Raw Crunchbase	17.26%	Primary predictor
funding_rounds	Integer	Raw Crunchbase	0%	Count predictor
market	String	Raw Crunchbase	8.03%	Self-reported sector
country_code	String	Raw Crunchbase	10.67%	ISO 3-letter
founded_year	Float	Parsed from founded_at	varies	Year dimension
is_closed	Binary	1 if status=='closed'	0%	Target variable
status_group	String	Mapped active/terminal/other	0%	Simplified grouping
funding_duration_days	Integer	last − first funding	varies	Engagement duration
reached_series_a	Binary	1 if round_A > 0	0%	Milestone indicator
log_funding_total	Float	log(1 + funding)	0%	Regression normalization
days_to_first_funding	Integer	first_funding − founded	varies	Speed-to-investor

B. Python ETL Code Snippets

Reference: `notebooks/02_cleaning.ipynb`

Listing 1: Key ETL Transformations

```python
# Step 3     Remove duplicate startups
df = df.drop_duplicates()
df = df.drop_duplicates(subset=['permalink'], keep='first')

# Step 5     Standardize numeric funding columns
for c in funding_cols:
    df[c] = pd.to_numeric(df[c], errors='coerce')
    df.loc[df[c] < 0, c] = np.nan

# Step 6     Engineer analysis features
df['is_closed'] = (df['status'] == 'closed').astype(int)
df['funding_duration_days'] = (
    pd.to_datetime(df['last_funding_at'])
    - pd.to_datetime(df['first_funding_at'])
).dt.days
df['reached_series_a'] = (df['round_A'] > 0).astype(int)
df['log_funding_total']  = np.log1p(df['funding_total_usd'])
```

Contribution Matrix

All claims are verifiable through GitHub Insights, pull request history, and committed files on the `dev` branch.

Table 10: Team Contribution Matrix

Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT
Riya Garg	**Owner**	Support	Support	Support	Support	Support	Support
Rashmi	Support	**Owner**	Support	Support	Support	Support	Support
Vachan Gupta	Support	Support	**Owner**	Support	Support	Support	Support
Vansh Panwar	Support	Support	Support	**Owner**	Support	Support	Support
Ujjwal Bharadwaj	Support	Support	Support	Support	**Owner**	Support	Support
Abhinav Choudhary	Support	Support	Support	Support	Support	**Owner**	**Owner**

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts. All analysis, code, and recommendations are the original work of Section C, Group 17.

Team Lead Signature: _____ **Date:** _____